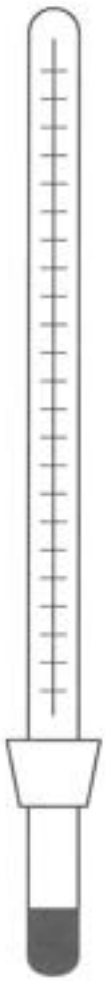
18 Ectotherms and endotherms

Multicellular animals can only survive within a relatively narrow band of temperatures. There are exceptions but the lower limit is usually 0°C (freezing point of water) and 45°C (when proteins are *denatured*, or 'cooked'). The optimum temperatures for particular organisms are even more restricted.



1 Explain what is meant by *optimum temperature*.

Cellular respiration releases energy when glucose is oxidised (combines with oxygen to form carbon dioxide and water). Most of this energy is released as heat and will affect the body temperature of the organism. We can divide animals into two groups, depending on how their body temperature is regulated – *ectotherms* and *endotherms*.

Ectotherms

Many animals lose most of the heat energy generated from cellular respiration to their environment. Their body temperature is variable and is determined by their surroundings. These animals are ectotherms. You may also see such animals referred to as being *poikilothermic* (having variable temperatures) or even (inaccurately) as being *cold-blooded*. Usually their body temperature is only slightly higher than ambient temperature but, depending on their activity level, can be much warmer than their surroundings.

2 Explain what is meant by *ambient temperature*.

Animals can absorb heat from the sun or hot surfaces (generally heated by the sun). They lose heat by radiation from their bodies, evaporation (perspiration etc) and conduction to cold surfaces. Water conducts heat better than air does, but large masses of water vary less in temperature than the atmosphere. This is because water has a greater heat capacity than air and takes longer to heat up and cool down. *Aquatic* ectotherms are well adapted to the temperature of their surroundings. *Terrestrial* ectotherms need to have greater responses to extremes of temperature.

3 Distinguish between the terms *aquatic* and *terrestrial*.

4 Suggest why terrestrial ectotherms must have greater responses to temperature extremes than aquatic ectotherms.

> *The metabolism of an organism is the sum of all of the chemical reactions occurring in that organism.*

The activity of ectotherms varies with temperature. The rate of their metabolism increases when temperatures rise. If temperatures are low, the rate of reactions (both those that break and make molecules) are slow and ectotherms are sluggish, ie move and respond slowly. When land-based ectotherms need to raise their temperature they expose more of their body to the sun ('sunbaking'). Lizards and butterflies commonly do this.

5 Name two Australian ectotherms that might sunbake.

If there are prolonged periods of cold weather then ectotherms can slow their metabolisms down considerably and become inactive. When reptiles do this they are said to be *torpid*. Insects have a similar process called *diapause*.

6 Name an Australian reptile that might become torpid.

7 Name an Australian insect that might go through diapause.



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The table below shows the relative metabolic rates at different temperatures for an organism.

Temperature (°C)	0	10	20	30	40
Metabolic rate (arbitrary units)	1	2	4	8	16

8 Draw a best fit line graph using these figures. (Draw temperature on x-axis.)

9 Describe the relationship between metabolic rates and temperature.

10 Explain why the temperature doesn't go past 40°C in this table.

11 Give an explanation as to why insects in alpine areas, as a rule, tend to be small and dark.

Endotherms

Some animals (mainly birds and mammals) have mechanisms that cut down loss of heat energy to surroundings. They can use the energy generated by cell respiration to keep a relatively steady body temperature. They are also called *homoeothermic* (uniform temperature) or (inaccurately) *warm-blooded*.

12 Birds have feathers as insulation to minimise heat loss. Name two other specific animal adaptations that reduce heat loss.

When conditions become too cold, some endotherms hibernate. The animal seeks out a sheltered place. Its metabolism slows down and body temperature drops. The mountain pygmy-possum hibernates below a layer of insulating snow. Some frogs and lizards (which are ectotherms) use a similar strategy.

The metabolic rate of endotherms is higher than that of ectotherms and they remain active over a wider range of temperatures. The higher rate of metabolism means that endotherms use more energy and hence need to consume more food than an ectotherm of equivalent size.

13 Outline why it might be an advantage for an endotherm to be more active than an ectotherm.

Endotherms that live in hot areas, for example deserts, have techniques to keep cool. They can burrow underground, become nocturnal or minimise their activity during hot spells.

14 Define *nocturnal*.

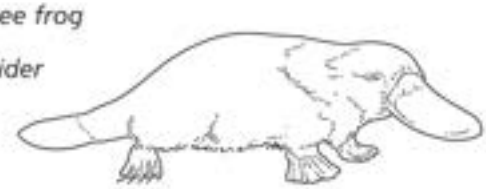
Evaporation of water from bodies by sweating or panting can be used to keep animals cool. This can cause problems for animals living in hot areas where water supplies are scarce. Blood is an important way of moving heat around the body. When heat needs to be radiated from the body, blood flow near the surface is increased. When conditions are cold and heat loss needs to be minimised, blood flow near the surface of the skin is reduced.

15 Explain why a red kangaroo might lick the inside of its paws when temperatures are high.

16 What is the advantage for an animal such a rabbit-eared bandicoot to have large ears?

17 Divide the following Australian animals into ectotherms and endotherms:

barramundi *bats* *blue-tongued lizard* *Bogong moth* *echidna*
emu *frill-necked lizard* *goanna* *green tree frog*
platypus *shark* *sugar glider* *white tip spider*



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18 Endotherms (if sufficiently insulated) can cope better with conditions much colder than their body temperature than they can with conditions above their body temperature. Give a possible reason for this.

19 Bees are ectotherms. In the winter, if the temperature in a hive falls below a certain point, the bees become very active. In the summer, if the hive becomes too warm the bees beat their wings much more vigorously than normal. Give a possible reason for each of these two occurrences.

What do the following statements define?

20 *'A state in which the body temperature, heart rate and respiration of an animal fall well below normal'.*

21 *'All of the chemical processes taking part in the body, including those that break down large molecules and those that make large molecules'.*

Decide whether the following statements (22–31) are true or false:

22 Aquatic ectotherms must be able to respond to greater temperature extremes than terrestrial ectotherms. _____

23 Ectotherms obtain most of their body heat from their surroundings. _____

24 Invertebrates and fish, reptiles and amphibians are endothermic. _____

25 *Torpid* and *diapause* refer to periods of increased activity. _____

26 CO_2 and H_2O are products of cell respiration. _____

27 Multicellular animals can only survive within a narrow temperature range. _____

28 Very little of the energy produced during respiration is released as heat. _____

29 Ectotherms are also called poikilotherms. _____

30 Hibernation can occur within both endothermic and ectothermic species. _____

31 Endotherms obtain most of their body heat from their cell metabolism. _____

32 Define *hibernate* and give some of its benefits.

33 Animals that live in cold climates (eg the arctic) often have a layer of insulating fat (adipose tissue). Name two examples of these types of animals.

34 Describe how humans can deal with hot conditions using technology.

35 Bees live in hives. Identify another type of insect that lives in similar conditions.

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